

Howard Tang

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EDUCATION

Princeton University

B.S.E. in Electrical and Computer Engineering

Relevant Coursework: Contemporary Logic Design, Data Structures and Algorithms, Machine Learning for Predictive Data Analytics

Princeton, NJ · Expected May 2029

GPA: 3.78 / 4.00

RESEARCH & WORK EXPERIENCE

National Center for Supercomputing Applications (NCSA) · Research Intern

Urbana-Champaign, IL · May – July 2026

- Engineered a human-centric 4D scene reconstruction pipeline by integrating SMPL-X parametric body models into an existing camera-redirection framework, resolving the system's core failure mode on inputs with complex human motion, self-occlusion, and out-of-frame subjects.
- Designed a joint-correspondence alignment method (ViTPose++ 2D pose estimation with RANSAC-Umeyama registration) to register per-frame human mesh estimates into a global 3D scene space, replacing a less robust dense-correspondence approach and materially improving alignment accuracy.
- Built and refined a two-stage mesh canonicalization and temporal-smoothing algorithm (pelvis-centering, orientation normalization, MAD-based multi-pass outlier detection, quality-weighted quaternion averaging) to convert noisy framewise 3D pose estimates into geometrically consistent, temporally smooth human motion trajectories.
- Reduced inference time by 50% while achieving state-of-the-art-level 4D human-scene reconstruction and camera redirection.

Urban League of Rochester · Engineering Intern

Rochester, NY · June – July 2025

- Assembled and validated 12 quadcopters integrating flight controllers, ESCs, and RF communication systems, bringing all units to flight-ready status for a youth drone education program.
- Resolved hardware and software integration failures during flight testing, then authored technical documentation enabling instructors and students to operate systems independently without engineering support.

ROBOTICS & AUTONOMOUS SYSTEMS

Semi-Autonomous Robotic Vehicle · Designer & Builder

Rochester, NY · Sept. 2023 – March 2025

- Built a semi-autonomous differential-drive robot from scratch across 5 design cycles over 18 months, developing custom multi-layer PCBs and embedded C firmware to achieve a ~175 Hz real-time control loop with integrated motor control and sensor interfacing.
- Achieved odometry-based localization within ~3 mm over 1.5 m displacement tests and ~1 cm over 8 m competition runs by fusing IMU and encoder data via a complementary filter; improved obstacle-detection stopping reliability by fitting regression-based skid-distance curves to compensate for sensor-obstacle closing velocity.
- Coordinated mechanical, electrical, and software subsystem integration across all design cycles, performing structured subsystem validation and control tuning to improve autonomous reliability between iterations.
- Placed 3rd among 70+ teams at the 2025 NY Science Olympiad Robot Tours state competition, validated across full end-to-end autonomous runs.

HARDWARE DESIGN

Custom 16-Bit Processor · Contemporary Logic Design — Princeton University

Princeton, NJ · Fall 2025

- Designed and programmed 16-bit, 16-instruction processor in Verilog from scratch, validated via 100%-instruction-coverage testbenches and waveform analysis in Vivado across 70+ hours of hardware lab verification.

SOFTWARE & ALGORITHMS

Algorithms & Data Structures Projects · Data Structures and Algorithms — Princeton University

Princeton, NJ · Fall 2025

- Implemented a fraud-detection pipeline combining graph algorithms, MSTs, decision stumps, and a modified AdaBoost classifier to identify anomalous transaction patterns with measurable accuracy gains over baseline.
- Built algorithmic implementations including a seam-carving image compressor (dynamic programming + graph path search), a k-d tree for 2D spatial queries, and an autocomplete engine using efficient sorting and search algorithms.

Engineering Simulation Tools · Independent

Rochester, NY

- Developed 16+ Java tools spanning image segmentation, Fourier series, matrix transformations, and geometric modeling to support engineering design workflows and fabrication planning.

TECHNICAL SKILLS

Programming: Python, Java, Verilog, Embedded C/C++ (Arduino)

Robotics & Controls: PID control, sensor fusion, complementary filtering, odometry, differential-drive kinematics

Embedded & Hardware: PCB design, embedded firmware, RTL/FSM design (Vivado), SLURM/HPC

Computer Vision & ML: Video diffusion models (Wan2.2 VACE), 3D scene reconstruction, SMPL-X, VGGT/VGGT-Omega, SAM2.0

Engineering Design: CAD (Fusion 360, Onshape; 100+ designs), FDM 3D printing, laser cutting, soldering